

Tutorial 3.

Q1) What is ALOHA and what are its drawbacks? How CSMA/CD overcomes it? Explain.

→ [ALOHA is a simple communication protocol designed for a shared medium in computer networks. Devices transmit

⇒ ALOHA is a random access medium access control (MAC) protocol used in shared communication channels. In ALOHA a station transmits data whenever it has data to send without checking whether the channel is free or busy.

⇒ If two or more stations transmit simultaneously, a collision occurs and the transmitted frames are destroyed. The station then waits for a random amount of time before retransmitting.

⇒ Types of ALOHA.

1. → pure ALOHA

- Stations transmit at any time.
- Maximum efficiency $\approx 18.4\%$.

2. → Slotted ALOHA.

- Time is divided into slots.
- Transmission can begin only at the start of the slot.
- Maximum efficiency $\approx 36.8\%$.

⇒ Drawback of ALOHA:-

1> High collision Rate:-

- Device transmit without checking if the channel is busy, causing frequent collisions and loss of data.

2> low efficiency.

- pure ALOHA :- 18.74% max channel utilization.
- Slotted ALOHA :- 36.8% max channel utilization.
- this limits performance in high-traffic scenarios.

3> Unpredictable Delays.

- collisions leads to random retransmissions, making delays variable and unsuitable for real time communication.

4> Poor Scalability:-

- performance degrades significantly as more devices share the channel.

5> No channel sensing:-

- transmissions occurs blindly, increasing the likelihood of collisions.

##

⇒ CSMA/CD (Carrier Sense Multiple Access with collision detection) improves ALOHA by making stations listen before transmitting.

Working principle:-

1. Carrier sense

- A station checks whether the channel is idle or busy.

2. Multiple Access.

- many stations share the same communication medium.

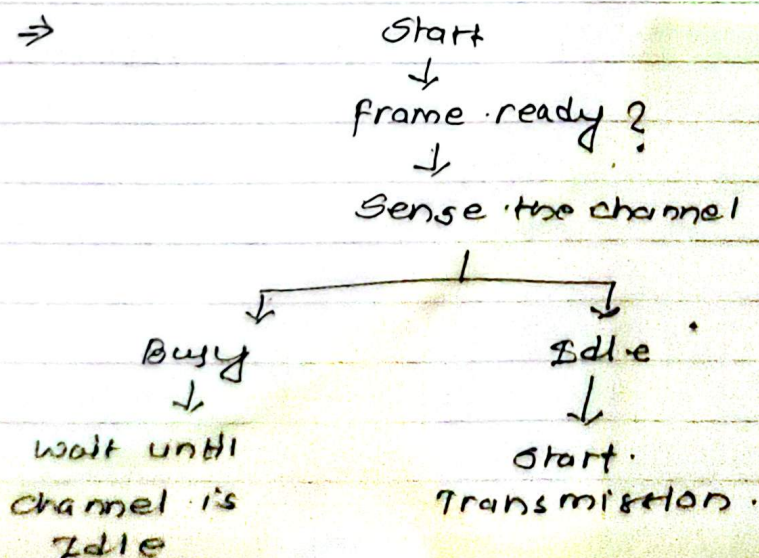
3. Collision Detection.

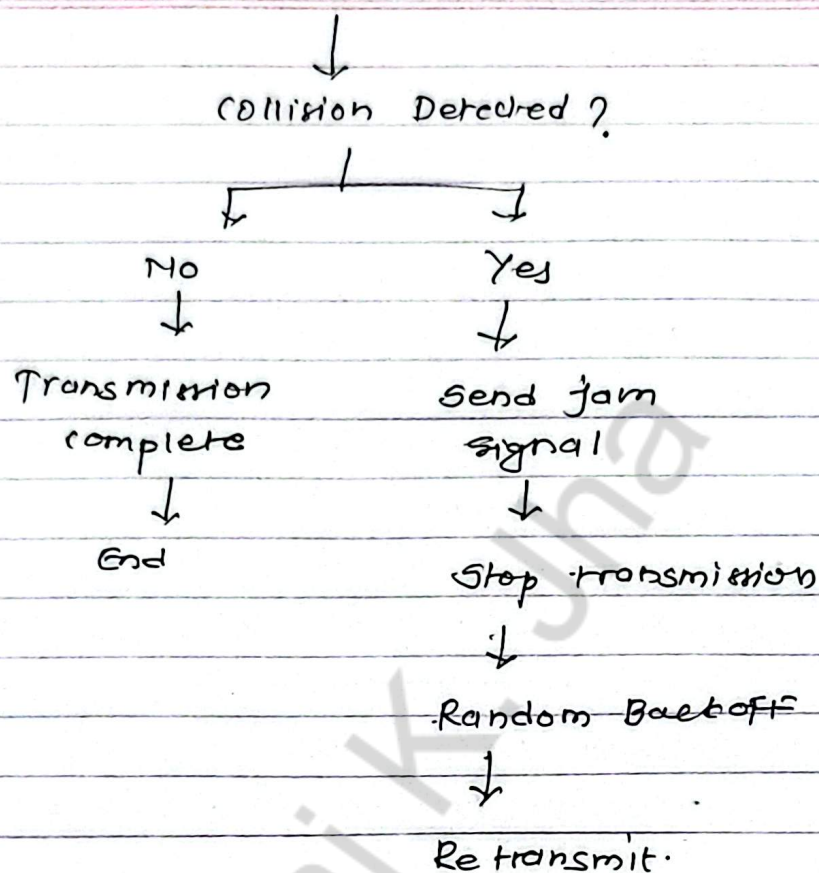
- While transmitting, the station monitors the channel.
- If a collision is detected, transmission is stopped immediately.

4. Backoff Algorithm.

- Stations wait for a random time before retransmitting.

⇒





- ⇒ CSMA/CD overcomes these drawbacks by sensing the channel before transmission and detecting collision during transmission.
- When a collision occurs, the station sends a jam signal, stop transmission, waits for a random backoff time & retransmits the frame.
- This reduces collisions; improves bandwidth utilization, & increases n/w efficiency.

2. Differentiate between OSI and TCP/IP Model.

→ The OSI (Open System Interconnection) model and TCP/IP (Transmission Control Protocol / Internet Protocol) models are networking architecture used to describe how data is transmitted over a network. OSI is a reference model whereas TCP/IP is a practical model used on the Internet.

⇒ OSI Model	TCP/IP Model.
- Developed by ISO (International Organization for Standardization) - Consists of 7 layers. - It is a reference model - Layers are clearly separated - Developed before protocols were defined - Mainly used for understanding and teaching n/w concepts. - N/w, Transport, & Application services are strictly distinguished - More complex	- Developed by DARPA Department of Defense, USA. - Consists of 4 layers. - It is a protocol suite/model - Some layers are combined. - Protocols exist before the models. - Widely used in real-world internet communication. - Less strict separation b/w layers. - Simpler & more practical

OSI Model

Tcp/Ip Model

Application
presentation
session



Application-

Transport



Transport.

N/w



Internet.

Data link
physical



Network Access.

PDU : protocol Data unit .

3. OSI layer Architecture is a Reference Model. Support this statement and Explain the basic function of each layer.

→ The OSI model is called reference model because it provides a standard framework for understanding and designing network communication.

→ It does not define specific protocols, instead it describes how different networking functions should be organized into layers.

→ It serves as a guideline for hardware manufacturers, software developers and also engineers to ensure interoperability between different systems.

⇒ Functions of each OSI layer .

↳ physical layer .

- Transmits raw bit over the communication medium .

- Defines cables, connectors, voltage levels, and data rates .

- eg: Ethernet cable,
Fiber optic cables,
Hubs .

 PDU = Bits

2. Data link layer.

- provides nodes to node delivery.
- Detects and corrects transmission errors.
- control access to the shared medium.

PDU:- frame.

⇒ sublayers.

LLC (Logical Link Control)

MAC (Media Access Control).

⇒ eg:- switches, MAC Address.

3. Network layer.

- provides logical addressing & routing.
- determines the best path for data transmission.

PDU:- packet.

eg:- IP Address, Routers.

4. Transport layer.

- provides end to end communication.
- ensures reliable data delivery.
- performs flow control & error recovery.

PDU:- Segment.

eg:- TCP, UDP.

6> Session layer.

- Establishes, manages & terminates communication sessions.
- Synchronizes communication b/w applications.

eg:-

Remote login sessions.
Video conferencing sessions.

6> presentation layer.

- Translates data formats.
- performs encryption & decryption.
- compresses and decompresses data.

eg:-

JPEG, GIF, SSL/TLS.

7> Application layer.

- provides n/w services directly to users.
- Acts as the interface b/w users & the n/w.

eg:-

HTTP, FTP, SMTP, DNS.

	layer
Application.	7
presentation	6
Session.	5
Transport.	4
Network	3
data link	2
physical	1.

4. Write the advantage of layered network model.

→ A layered network model divides the communication process into different layers, where each layer performs specific functions and provides services to the layer above it. Examples include the OSI and TCP/IP models.

⇒ Advantages of Layered Network Model.

1. Simplified Network Design:-

- Complex networking tasks are divided into smaller, manageable layers.
- Makes network design easier and more organized.

2. Easy Troubleshooting.

- Problems can be isolated to a specific layer.
- Reduces the time required to identify & fix faults.

3. Modularity:-

- Each layer operates independently.
- Changes in one layer do not significantly affect other layers.

4. Standardization

- Provides standard rules and protocols for communication.
- Ensures compatibility between devices from different manufacturers.

5) Interoperability:-

- Different hardware and software systems can communicate effectively
- promotes vendor-independent networking.

6) Easier Maintenance and updates:-

- New technologies can be introduced in one layer without redesigning the entire network.
- Simplifies network expansion and updates.

7) Reusability:-

- Services of one layer can be ^{re}used by multiple higher layers.
- Reduces duplication of functions.

8) Flexibility:-

- Supports multiple protocols and technologies within the same architecture.
- Allows networks to adapt to changing requirements.

- 5> Explain the concept of addressing in each layer of TCP/IP model.
- In the TCP/IP model, each layer uses a different type of addressing to ensure proper delivery of data from a source to a destination.

⇒ Addressing in TCP/IP layers.

1> Applications layer → Name Addressing concept:-

- use human-readable names to identify services or resources.

eg:-

Domain names like:- `www.google.com`.

purpose:-

- Easy for humans to remember instead of numerical addresses.

2> Transport layer → port Addressing concept:-

- use port numbers to identify specific processes or applications on a device.

- eg:-

HTTP → port 80, HTTPS → port 443

purpose:-

- Ensure correct delivery of data to the right application.

Address Type:- port no (16-bit).

Address Type:- logical Address.

3.) Internet layer → logical Addressing (IP Address).

concept:-

- Use IP address to identify devices across networks.

e.g:-

IPv4:- 192.168.1.1

IPv6:- 2001:db8::1

purpose:-

- used for routing packets between different n/w.

4.) Network Access layer → physical Addressing (MAC address).

concept:-

- use ~~headphones~~ hardware address assigned to network interface card (NIC)

e.g:- 00:1A:2B:3C:4D:5E

purpose:-

- Used for communication with in the same local network (LAN)

Address Type:- MAC Address (48-bit).

Q3 Define X.25, Frame Relay and ATM briefly.

- These are WAN (Wide Area Network) switching technologies, used to transmit data over long distance.
- They mainly differ in how they handle connection, error control, and speed.

Q4 X.25.

- X.25 is an old packet-switched WAN protocol designed for reliable data communication over noisy analog lines.

→ Key features:-

- connection-oriented protocol.
- provides error detection and correction at each hop.
- use virtual circuits.
- works well in unreliable networks.

→ Characteristics:-

- High overhead due to error checking.
- slower compared to modern technologies.
- Reliable but inefficient.

→ Uses:-

- Early public data network.
- Banking and terminal systems (historically).

B. > Frame Relay

- frame Relay is a high speed packet switched WAN technology that is simpler and faster than X.25.

→ Key Features:-

- Connection-oriented (uses virtual ckt's)
- Minimal error controls (handled by end devices).
- Faster than X.25.
- Reduced overhead.

→ Characteristics:-

- Efficient for bursty traffic
- Lower reliability compared to X.25.
- Uses DLCI (Data Link Connection Identifier).

→ Uses:-

- Corporate WANs (older Networks)
- LAN interconnection over WAN.

C. > ATM.

- ATM is a cell-based switching technology that transmits data in fixed size cell (53 bytes).

→ Key Features:-

- Uses fixed size cell (53 bytes).
- Supports both voice, video, and data.
- High-speed transmission.
- Connection oriented.

→ Characteristics:-

- very fast and efficient.
- suitable for real time application.
- complex and expensive.

→ usage:-

- Telecom networks.
- Backbone networks.
- multimedia communication.

⇒ What is PDU?

→ protocol data unit is the unit of data exchanged between peer layers in a network model (such as OSI or TCP/IP).

In other word:-

- When data moves down or up the network layers, each layer adds its own information. The resulting formatted data at each layer is called a PDU.

→ It represents data along with control information (headers and sometimes trailers) added by each layer during transmission.

PDU at Different OSI layers.

OSI layer

PDU Name:-

- | | |
|------------------------------------|----------------------------------|
| - Application/presentation/session | data. |
| - transport layer | segment (TCP) / datagram
UDP. |
| - Network layer | packets. |
| - data link layer | frame. |
| - physical layer | bits. |

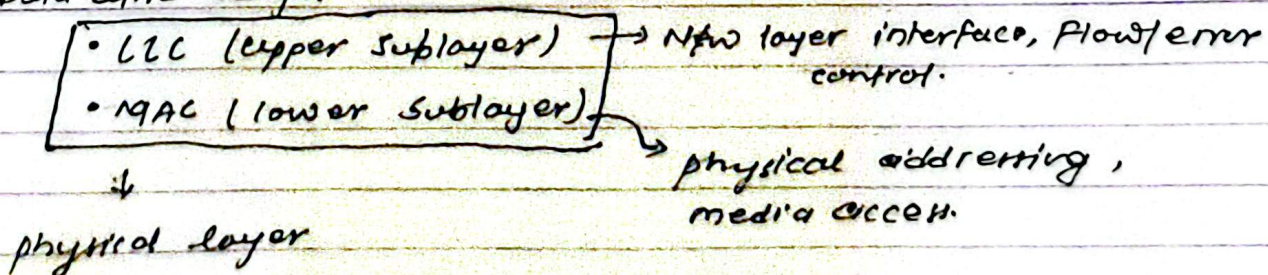
8. Differentiate between LLC and MAC sublayer of OSI reference model.

→ In the data link layer (layer 2) of the OSI model, there are two sublayers:-

- LLC (Logical Link control)
- MAC (Media Access control.)

→ Both work together to ensure reliable communication over a network, but they perform different roles.

Data Link Layer.



Features	LLC	MAC
Full form:-	- Logical Link Control	- Media Access Control
Main function:-	- provides interface b/w Network layer & Data link layer	- controls access to the physical transmission medium
Responsibility:-	- flow control, error control, and protocol multiplexing	- physical addressing and channel access control
Address used:-	- No physical address	- Uses MAC address
position in layer	- Upper sublayer	- lower sublayer
Standards:-	IEEE 802.2	IEEE 802.3/802.11
Error handling:-	provides error control (basic)	No error control (handled by LLC or higher layer).
Medium Access	Does not handle medium access	Determines when a device can transmit.

9. Explain different types of signal impairments in communication systems.

→ In a communication system, when a signal travels from sender to receiver, it gets degraded or distorted due to various factors. These problems are called signal impairments.

1. Attenuation.

- Attenuation is the loss of signal strength as it travels through a medium.
- Causes:-
 - Resistance of transmission medium.
 - Distance between sender and receiver.
- Effect:-
 - Signal becomes weak.
 - Receivers may not detect signal properly.
- Solution:-
 - Use repeaters or amplifiers.

2. Distortion:-

- Distortion occurs when the shape of the signal changes during transmission.
- Causes:-
 - Different frequency components travel at different speeds.
 - Delay in signal components.
- Effects:-
 - Output signal becomes different from input signal.
- Solution:-
 - Use equalizers and proper filtering.

3> Noise:-

- Noise is an unwanted random signal that interferes with the original signal.

→ Types of Noise:-

- Thermal Noise
- crosstalk.
- Intermodulation noise
- impulse noise.

→ Effect:-

- Reduces clarity of signal
- causes errors in data transmission.

→ Solution:-

- Shielding cables.
- using error detection / correction techniques.

10) What do you mean by propagation? What do you need it? Explain different propagation mechanisms.

→ Propagation is the process by which an electromagnetic signal (radio wave) travels from a transmitter to a receiver through a communication medium or free space.

→ In wireless communication, information is transmitted through electromagnetic waves and propagation describes how these waves move through the environment.

⇒ Propagation^{is} Needed because:-

- It enables wireless communication.
- It allows signals to travel from sender to receiver over long distances.
- It supports communication systems such as radio, television, mobile networks, satellite communication & wifi.
- It helps determine the coverage area and quality of communication.



without propagation, wireless transmission would not be possible.

⇒ Different propagation mechanisms are explained below:-
↳ When electromagnetic waves travel from transmitter to receiver, they can propagate in different ways.

1.) Ground Wave propagation (surface wave).

- ⇒
- Radio wave travel along the surface of the earth.
 - Waves follow the curvature of the Earth.

⇒ Frequency range:- upto about 2 MHz.

⇒ Application:-

- AM radio broadcasting
- Marine communication.
- Navigation system.

⇒ Advantages.

- can cover long distance.
- Not affected much by obstacles.

2.) Sky wave propagation.

⇒ Radio wave are transmitted toward the ionosphere

⇒ The ionosphere reflects/refracts the waves back to earth.

⇒ Frequency Range:- Approximately 2 MHz to 30 MHz

⇒ Application:-

- International radio broadcasting

- Amateur (ham) radio communication.
- Long distance communication.

⇒ Advantage:-

- Enables communication beyond the horizon.

3.→ Line of sight (LOS) propagation.

- concept:-

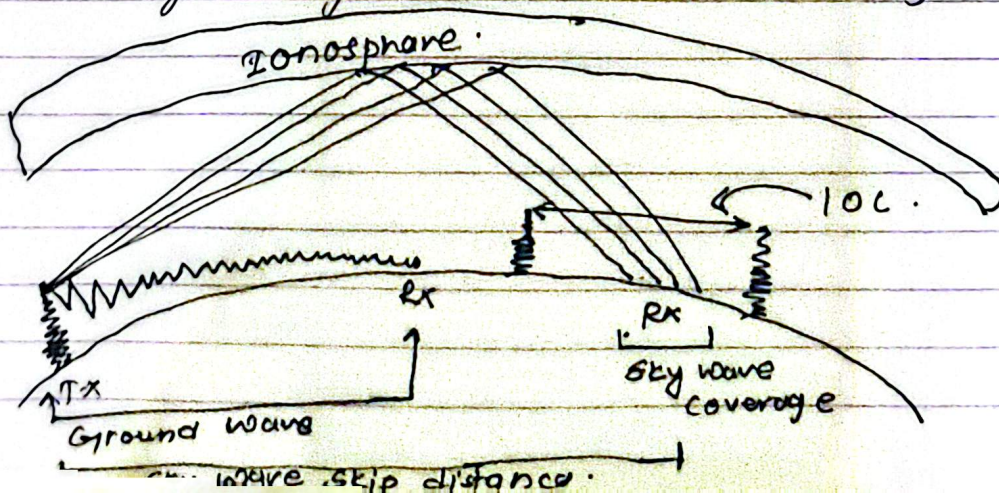
- Signals travel directly from transmitter to receiver.
- Requires an unobstructed path.

⇒ Frequency Range:- Above 30 MHz.

⇒ Applications:-

- Television broadcasting
- Mobile communication.
- Wifi
- Microwave links.
- Satellite communications.

⇒ Advantage:- High data rates and better signal quality.



12. Explain the working principle of optical fiber. What is the importance of cladding in optical fiber?

→ Optical Fiber:-

- It is a guided transmission medium that transmits information in the form of light signals through a thin strand of glass or plastic.

⇒ Working principle of optical fiber:-

↳ Optical fiber works on the principle of -

- Total Internal Reflection (TIR).

- When light travels from a denser medium (core) to a rarer medium (cladding) at an angle greater than the critical angle, it is completely reflected back into the core.
- As a result, the light signal continuously bounces inside the fiber core and travels long distance with very little loss.

⇒ Working Steps:-

1. A transmitter converts electrical signals into light signal.
2. The light enters the fiber core.
3. The light undergoes repeated TIR.
4. The light travels through the fiber with minimal attenuation.
5. At the receiving end, the light is converted back into the electrical signal.

⇒ Importance of Cladding in optical fiber.

- The cladding is a layer of material surrounding the core with a lower refractive index than the core.

1. Enables Total Internal Reflection.

↳ keeps light confined within the core.

↳ prevents light from escaping.

2. Reduces signal loss.

↳ minimizes attenuation.

↳ Improves transmission efficiency.

3. protects the core.

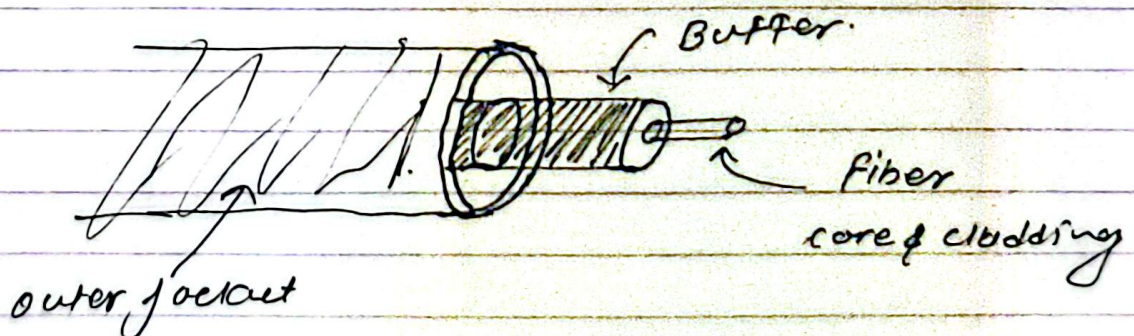
↳ provides mechanical protection to the core

4. Improves signal quality.

↳ prevents distortion & leakage of light.

5. Increases Transmission Distance.

↳ Allows signals to travel longer distance without significant loss.



13. Compare Guided Transmission media in terms of Bandwidth, Channel capacity, SNR, security, cost Attenuation, delay and Repeater spacing

→ Guided transmission media are communication channels in which signals travel through a physical path.

- The main types are:-

1. Twisted pair cable

2. coaxial cable

3. Optical fiber cable

⇒ parameter	Twisted pair cables	coaxial cable	optical fiber.
Bandwidth	low	medium	very high.
channel capacity	L	M	V.H.
SNR	L	M	V.H.
Security	L	M	V.H.
Cost	L	M	H
Attenuation	H	M	V.L
propagation Delay	Higher	M	V.L
Repeater spacing	short (2-3)	medium (1-5 km)	long (50-100 km)
Data Rate	L	M	V.H

14) Explain the electromagnetic spectrum and its Applications in wireless transmission media with their mode of propagation.

→ The Electromagnetic spectrum is the complete range of electromagnetic waves arranged according to their frequencies or wavelengths.

→ Wireless communication uses different portions of the electromagnetic spectrum to transmit information through free space without physical cables.

⇒ Application in wireless Transmission:-

Spectrum Band	Frequency Range	Application.
• Radio wave	3Hz → 1GHz	AM/FM radio, TV broadcasting
• Microwaves	1GHz → 300GHz	Mobile communication, wifi-satellite
• Infrared	300GHz - 400THz	Remote control, short range communication.
• Visible light	400THz - 790THz	Fiber optics, optical communication.

15. Explain the working principle of satellite communication.

→ satellite communication is a method of communication in which an artificial satellite acts as a repeater (relay station) in space.

→ It receives signals from an Earth station, amplifies them, and retransmits them to another Earth station.

⇒ Working principles of satellite communication.

→ The working of satellite communication involves three main components.

1. Transmitting Earth station.

2. Satellite (Transponder).

3. Receiving Earth station.

Steps of communication.

1. Uplink Transmission.

- The transmitting Earth station sends signals to the satellite.

- This transmission path is called the uplink.

- Uplink frequency is generally higher than the downlink frequency.

2.) Signal processing by satellite.

- The satellite receives the signals through its antenna.
- The transponder amplifies, filters, and changes the frequency of the signal to avoid interferences.

3.) Downlink Transmission.

- The processed signal is transmitted back to the earth.
- This transmission path is called the 'downlink'.
- The receiving Earth station receives and decodes the signal.